



OBSIDIAN '26:

Smart QR Entry & Access

Management System

Objective: To replace manual guest verification at the Freshers venue with a secure, mobile-first digital entry system that validates registered junior IDs from an uploaded Excel/CSV list, generates the official event QR code, records check-ins in real time, and delivers an immersive 3D OBSIDIAN-themed entry experience.

1. Target Users

- **Junior Attendees / Venue Visitors:** Arrive at the Freshers venue, scan the event QR code, enter their Student/College ID, and receive an instant access result.
- **Event Organizer/Admin Team:** Authorized members of my team who can access the registration data, manage the Excel/CSV list and venue QR, monitor check-ins, search students, and export attendance records.

2. Core System Pages

Page 1: QR Landing & 3D OBSIDIAN Intro

- **Purpose:** Create the first immersive interaction after a junior scans the venue QR.
- **Features:** Mobile-first landing page, animated 3D obsidian/amethyst crystal, black-purple contour background, subtle particles, OBSIDIAN '26 branding, and *"Unfold the Unknown"*.
- **Performance:** Lazy-load the 3D scene and provide a lightweight fallback on low-end devices.

Page 2: Student ID Verification

- **Purpose:** Verify whether the attendee is in the official registered-student list.
- **Input:** Student/College ID with trim, case normalization, and format validation.
- **Logic:** Search the backend database by unique Student ID; never expose the complete Excel/student list in the browser.
- **UX:** Show a short themed 'Scanning / Verifying Registration' animation during the lookup.

Page 3: Access Result & Check-In

- **Registered + Not Checked In:** Show **ACCESS GRANTED**, selected student details, and record the check-in time.
- **Registered + Already Checked In:** Show **ALREADY CHECKED IN** with the previous entry time; do not duplicate the check-in.
- **ID Not Found:** Show **ACCESS DENIED** and direct the student to the registration desk.
- **Visual Feedback:** Purple/white crystal glow for success and controlled deep-red/crimson effects for denial.

Page 4: Admin Login & Authentication

- **Purpose:** Protect organizer-only pages and event data.
- **Features:** Secure login, session handling, logout, protected admin routes, and Admin vs. Volunteer/Entry-Desk permissions.

Page 5: Event Command Center (Admin Dashboard)

- **Purpose:** Give organizers an at-a-glance view of event entry status.
- **KPIs:** Total Registered, Checked In, Not Arrived, and Denied/Invalid Attempts.
- **Features:** Recent check-ins, live search, department/branch filter, status filter, and timestamps.

Page 6: Student Registry & Excel/CSV Import

- **Purpose:** Convert the existing form-response spreadsheet into the verification database.
- **Features:** Upload .xlsx/.xls/.csv, preview rows, map Student ID, validate required fields, and report invalid/duplicate records.
- **Data:** Student ID as a unique key plus Name, Mobile, Department/Branch, Email, and other imported registration fields.
- **Safety:** Import on the server; personal data is visible only to authorized admins.

Page 7: Event QR Code Management

- **Purpose:** Generate the QR code printed/displayed at the Freshers venue.
- **Features:** QR linked to the public entry URL, preview, regenerate, and download in high-quality PNG/SVG/PDF-ready form.
- **Security Option:** Event token / rotating token mode to reduce reuse of a forwarded QR screenshot.

Page 8: Attendance, Audit & Reports

- **Purpose:** Maintain the final attendance record and organizer audit trail.
 - **Features:** Registered-student table with check-in status/time, search/filter, admin correction, and attempt history.
 - **Exports:** One-click CSV/Excel/PDF for Checked In, Not Arrived, and full attendance.
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3. Logic & Workflow Summary

- 1 **Registration Import:** Admin uploads Excel/CSV; the system validates it and stores each Student ID as a unique attendee.
- 2 **QR Setup:** Admin generates the OBSIDIAN '26 entry QR and places it at the venue.
- 3 **Scan:** Junior scans the QR and opens the themed 3D landing page.
- 4 **Verification:** Student enters their ID; the backend securely searches the database.
- 5 **Decision:** Found + not checked in -> **Access Granted**; found + checked in -> **Already Checked In**; not found -> **Access Denied**.
- 6 **Live Update:** Successful entry stores date/time and updates the dashboard.
- 7 **Post-Event:** Admin exports attendance and audit reports.

4. Technical Requirements

- **Frontend:** Next.js + React + TypeScript + Tailwind CSS; responsive, mobile-first UI.
- **3D / Motion:** Three.js / React Three Fiber with optimized crystal, particles, contour effects, and access transitions.
- **Backend:** Next.js API routes/server actions with input validation, rate limiting, and server-side verification.
- **Database & Auth:** Supabase/PostgreSQL for Students, Check-Ins, Event Settings, Admin Users, and Audit Logs.
- **Spreadsheet Import:** SheetJS/xlsx for Excel/CSV parsing and validation.
- **QR:** Reliable QR library; downloadable venue QR connected to the live event URL.
- **Security:** Server-side secrets, protected admin routes/RLS, sanitized IDs, duplicate-check prevention, and no public attendee dataset.
- **Deployment:** Vercel + Supabase over HTTPS using environment variables.
- **Design:** Official OBSIDIAN '26 black/deep-purple palette, faceted crystal centerpiece, topographic lines, cinematic look, and "*Unfold the Unknown*".

Design Reference: Official OBSIDIAN '26 Freshers 2K26 poster supplied with this task.